3 Figure 3.1 shows a polar bear and a person standing on thick ice. The figures are drawn to a scale of 1 cm : 100 cm

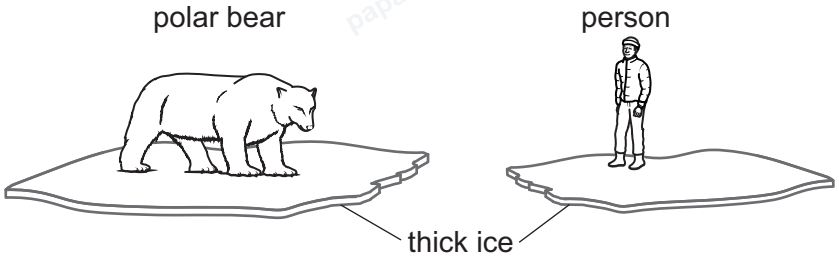


Figure 3.1

Table 3.1 contains information about the polar bear and the person.

Table 3.1

	polar bear	**person**
mass / kg	590	84
area in contact with ice / m^2	1.12	0.020

(a) (i) Calculate the weight of the polar bear.

weight = ... N [1]

(ii) Calculate the pressure that the polar bear exerts on the ice.

pressure = ... Pa [2]

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(iii) Use the numbers to complete each equation.

Each number may be used once, more than once or not at all.

0.125 7.0 8.0 56 392

mass of polar bear = ... × mass of person

area of polar bear in contact with ice = ... × area of person in contact with ice

pressure exerted by person = ... × pressure exerted by polar bear

[2]

(b) Under the ice is sea water with a density of 1030 kg / m^3.

Calculate the change in pressure Δp between the top of the sea water and a depth of 2.50 m.

Δp = ... Pa [2]

[Total: 7]

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